

RESEARCH INTERESTS

Natural Language Processing

Summarization, Generation, Low-Resource Languages

Generative AI

Large Language Models, Multimodal Foundation Models, Agentic AI

AI4Science

Healthcare, Linguistics, Sensing, Education

ACADEMIC EMPLOYMENT

University of North Carolina at Charlotte

Tenure-track Assistant Professor, School of Data Science

Aug. 2026 - Present

University of Hawaii at Manoa

Tenure-track Assistant Professor, Department of Information and Computer Sciences

Aug. 2024 - Jul. 2026

EDUCATION

University of California, Davis

Ph.D. in Computer Science

Thesis: Building Intelligent and Reliable Summarization Systems

Jun. 2024

Advisor: Dr. Jiawei Zhang

Georgia Institute of Technology

M.S. in Computational Science and Engineering

M.S. in Electrical and Computer Engineering

Dec. 2018

University of Illinois Urbana-Champaign

B.E. in Electrical Engineering

May 2016

INDUSTRY EXPERIENCE

Amazon Web Services (AWS) AI Lab

Applied Scientist Intern

Santa Clara, CA

Jun. 2023 - Sep. 2023

Megagon Labs

Research Intern

Mountain View, CA

Mar. 2023 - Jun. 2023

Tencent AI Lab

Research Intern

Seattle, WA

Jun. 2022 - Sep. 2022

Salesforce Research

Research Intern

Remote

May 2021 - Aug. 2021

GRANT AND PROJECT FUNDING

- **Key Personnel, NOAA CIFARM** 2026–2029
 - Project: *Cooperative Institute Fostering Aquaculture Research and Markets*
 - Award Amount: \$13.5M (University of Hawai'i Share: \$1.73M; Personal Share: \$460K)
- **Site PI, NSF Division of Information and Intelligent Systems (IIS)** 2024–2028
 - Project: *SCH: Personalized Machine Learning for Repeat Adverse Health Events using Novel Multimodal Self-Supervised Pretraining Methods*
 - Award Amount: \$1.2M (Personal Share: \$450K)
- **co-I, American Medical Association ChangeMedEd** 2025 - 2030
 - Project: *Developing Culturally Responsive AI for Rural Health: Utilizing Precision Education and Precision Coaching to Strengthen Hawaii's Rural Physician Workforce*
 - Award Amount: \$1.1M
- **Sole PI, OpenAI Researcher Access Program** 2024 - 2025
 - Computing Credits: \$7,500
- **Sole PI, Google Cloud Research Credits Program** 2024 - 2025
 - Computing Credits: \$5,000
- **Sole PI, NSF ACCESS Allocations Program** 2025
 - Project CIS250940: Exploring Multimodal Language Models for Wearable Signal Understanding and Prediction
 - Computing Credits: 1.5M
- **Sole PI, NSF ACCESS Allocations Program** 2024
 - Project CIS240616: Exploring fine-grained domain adaptation for text summarization
 - Computing Credits: 400K

AWARDS AND HONORS

- Best Contribution Award, EMNLP MRL workshop 2025
- UH Manoa Faculty Research Travel Award 2025
- UC Davis Graduate Group in Computer Science (GGCS) Research Fellowship 2023
- FSU Travel Award 2020
- Adelaide D. Wilson Graduate Fellowship 2019
- UIUC Russell E. Berthold Scholarship 2015

PUBLICATIONS

Citations: 1,583 (as of July 2026)

** indicates equal contributions — underscore indicates my Ph.D. students.*

Conference Proceeding

- [1] Haohan Yuan and **Haopeng Zhang**, “Understanding LLM reasoning for abstractive summarization”, in *Findings of the 2026 Conference of the Association for Computational Linguistics (ACL 2026)*, 2026.
- [2] Haohan Yuan, Sukhwa Hong, and **Haopeng Zhang**, “StrucSum: Graph-structured reasoning for long document extractive summarization with LLMs”, in *Findings of the Association for Computational Linguistics: EACL 2026*, 2026, pp. 3708–3721.
- [3] Zizhong Li, **Haopeng Zhang**, and Jiawei Zhang, “Token-level precise attack on RAG: Searching for the best alternatives to mislead generation”, in *Findings of the Association for Computational Linguistics: EACL 2026*, 2026, pp. 3193–3206.
- [4] Yizhu Wen, Nan Zhang, Haohan Yuan, Xun Chen, **Haopeng Zhang**, and Hanqing Guo, “Position: On the risks of generative engine optimization in the era of LLMs”, in *The Forty-Third International Conference on Machine Learning (ICML 2026)*, 2026.
- [5] Haohan Yuan and **Haopeng Zhang**, “DomainSum: A hierarchical benchmark for fine-grained domain shift in abstractive text summarization”, in *Findings of the Association for Computational Linguistics: NAACL*, Albuquerque, New Mexico, Apr. 2025, pp. 2219–2231.
- [6] Haohan Yuan, Siu Cheung Hui, and **Haopeng Zhang**, “A structure-aware generative model for biomedical event extraction”, in *International Conference on Database Systems for Advanced Applications*, Springer, 2025, pp. 257–272.
- [7] Zizhong Li, **Haopeng Zhang**, and Jiawei Zhang, “Learning by ranking: Data-efficient knowledge distillation from black-box LLMs for information retrieval”, in *2025 International Joint Conference on Neural Networks (IJCNN)*, IEEE, 2025, pp. 1–10.
- [8] Kaiying Lin, Hsiyu Chen, and **Haopeng Zhang**, “FormosanBench: Benchmarking low-resource austronesian languages in the era of large language models”, in *Findings of the Association for Computational Linguistics: EMNLP 2025*, Association for Computational Linguistics, Nov. 2025, pp. 16 527–16 539.
- [9] Yili Ren*, **Haopeng Zhang***, Haohan Yuan, Jingzhe Zhang, and Yitong Shen, “Wi-Chat: Large language model-powered Wi-Fi-based human activity recognition”, in *Proceedings of the International Workshop on Environmental Sensing Systems for Smart Cities*, 2025, pp. 8–13.
- [10] **Haopeng Zhang**, Hayate Iso, Sairam Gurajada, and Nikita Bhutani, “XATU: A fine-grained instruction-based benchmark for explainable text updates”, in *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING)*, Torino, Italia, May 2024, pp. 17 739–17 752.
- [11] Zizhong Li, **Haopeng Zhang**, and Jiawei Zhang, “Unveiling the magic: Investigating attention distillation in retrieval-augmented generation”, in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*, Mexico City, Mexico, Jun. 2024, pp. 745–754.
- [12] **Haopeng Zhang**, Sangwoo Cho, Kaiqiang Song, Xiaoyang Wang, Hongwei Wang, Jiawei Zhang, and Dong Yu, “Unsupervised multi-document summarization with holistic inference”, in *Findings of the Association for Computational Linguistics: IJCNLP-AACL*, Nusa Dua, Bali, Nov. 2023, pp. 123–133.
- [13] **Haopeng Zhang**, Xiao Liu, and Jiawei Zhang, “Extractive summarization via ChatGPT for faithful summary generation”, in *Findings of the Association for Computational Linguistics: EMNLP*, Singapore, Dec. 2023, pp. 3270–3278.
- [14] **Haopeng Zhang**, Xiao Liu, and Jiawei Zhang, “SummIt: Iterative text summarization via ChatGPT”, in *Findings of the Association for Computational Linguistics: EMNLP*, Singapore, Dec. 2023, pp. 10 644–10 657.

- [15] **Haopeng Zhang**, Xiao Liu*, and Jiawei Zhang, “DiffuSum: Generation enhanced extractive summarization with diffusion”, in *Findings of the Association for Computational Linguistics: ACL*, Toronto, Canada, Jul. 2023, pp. 13 089–13 100.
- [16] **Haopeng Zhang**, Xiao Liu, and Jiawei Zhang, “HEGEL: Hypergraph transformer for long document summarization”, in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Abu Dhabi, United Arab Emirates, Dec. 2022, pp. 10 167–10 176.
- [17] **Haopeng Zhang**, Semih Yavuz, Wojciech Kryscinski, Kazuma Hashimoto, and Yingbo Zhou, “Improving the faithfulness of abstractive summarization via entity coverage control”, in *Findings of the Association for Computational Linguistics: NAACL*, Seattle, United States, Jul. 2022, pp. 528–535.
- [18] **Haopeng Zhang** and Jiawei Zhang, “Text graph transformer for document classification”, in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Online, Nov. 2020, pp. 8322–8327.

Workshop Paper

- [19] Zizhong Li, **Haopeng Zhang**, and Jiawei Zhang, “MMViR: A multi-modal and multi-granularity representation for long-range video understanding”, in *1st International Workshop on Sustainable and Efficient Language, Vision, and Action Models (ACL SELVA)*, 2026.
- [20] **Haopeng Zhang**, “Bridging multimodal and video summarization: A unified survey”, in *Proceedings of The 5th New Frontiers in Summarization Workshop*, 2025, pp. 157–171.
- [21] Yiqiao Huang, Zhaorun Chen, He Qi, **Haopeng Zhang**, Hanchao Yu, and Zhuokai Zhao, “Autonomous multimodal reasoning via implicit chain-of-vision”, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2025, pp. 2988–2997.
- [22] Chengqi Zheng, Jianda Chen, Yueming Lyu, Wen Zheng Terence Ng, **Haopeng Zhang**, Yew-Soon Ong, Ivor Tsang, and Haiyan Yin, “MermaidFlow: Redefining agentic workflow generation via safety-constrained evolutionary programming”, in *Multi-Agent Systems in the Era of Foundation Models workshop at ICML (MAS)*, 2025.
- [23] **Haopeng Zhang**, Xiao Liu, and Jiawei Zhang, “Contrastive hierarchical discourse graph for scientific document summarization”, in *Proceedings of the 4th Workshop on Computational Approaches to Discourse at ACL*, Toronto, Canada, Jul. 2023, pp. 37–47.

Journal

- [24] Ali Kargarandehkordi, Aditi Jaiswal, Agnik Banerjee, Yang Qian, Christopher Slade, Yinan Sun, Tanvir Islam, Hiwot Tadesse, Alexander Kostrinsky-Thomas, Chanhyun Park, Urmimala Sarkar, Elaine Khoong, Nhung Nguyen, Xuhai Xu, Kristina Phillips, Roberto Benzo, **Haopeng Zhang**, Finale Doshi-Velez, and Peter Washington, “Personalised modeling of daily stress and blood pressure reactivity using wearable-based biosignals”, *npj Mental Health Research*, 2026.
- [25] **Haopeng Zhang**, Philip S. Yu, and Jiawei Zhang, “A systematic survey of text summarization: From statistical methods to large language models”, *ACM Computing Surveys*, vol. 57, no. 11, Jun. 2025.
- [26] Kaiying Lin, Abdur Rasool, Saimourya Surabhi, Cezmi Mutlu, **Haopeng Zhang**, Dennis Paul Wall, and Peter Washington, “Aiding LLMs with clinical scoresheets: No improvement in neurobehavioral diagnostic classification from text using basic prompting”, *JMIR AI*, 2025.
- [27] William O’Grady, **Haopeng Zhang**, and Miseon Lee, “The negative concord mystery: Insights from a language model”, *Information*, vol. 16, no. 8, p. 710, 2025.
- [28] **Haopeng Zhang**, “Building intelligent and reliable summarization systems”, Ph.D. dissertation, University of California, Davis, 2024.

Poster

- [29] Bardia Konh, Rex Imanaka, **Haopeng Zhang**, and Mitchell Kamrava, “Haptic simulation workstation for prostate brachytherapy training”, *Brachytherapy*, vol. 24, no. 4, S48, 2025.
- [30] Zizhong Li, **Haopeng Zhang**, and Jiawei Zhang, “Token-level precise attack on RAG: Searching for the best alternatives to mislead generation”, in *Proceedings of the 41st Annual Computer Security Applications Conference (ACSAC) Poster*, 2025.
- [31] Yili Ren*, **Haopeng Zhang***, Haohan Yuan, Jingzhe Zhang, and Yitong Shen, “Wi-Chat: Large Language Model Powered Wi-Fi Sensing”, in *Proceedings of the 23rd ACM International Conference on Mobile Systems, Applications, and Services (MobiSys) Posters*, ACM, 2025.

Preprint

- [32] Ethan Chung, Chuanjun Zheng, Jasper Tan, Jingxi Li, **Haopeng Zhang**, and Huaijin Chen, “Do AI understand imaging? a systematic benchmark of VLMs for computational imaging tasks”, *Manuscript*, 2026.
- [33] Mengqi Shi and **Haopeng Zhang**, “HAVEN: Hierarchically aligned multimodal benchmark for unified video understanding”, *arXiv preprint arXiv:2605.19223*, 2026.
- [34] Shouju Wang and **Haopeng Zhang**, “MPCI-Bench: A benchmark for multimodal pairwise contextual integrity evaluation of language model agents”, *arXiv preprint*, 2026.
- [35] MRL 25 authors, “Global PIQA: Evaluating physical commonsense reasoning across 100+ languages and cultures”, *arXiv preprint*, 2025.
- [36] Zizhong Li, **Haopeng Zhang**, and Jiawei Zhang, “A revisit of fake news dataset with augmented fact-checking by ChatGPT”, *arXiv preprint*, 2023.
- [37] Jiawei Zhang, **Haopeng Zhang**, Li Sun, and Congying Xia, “Graph-BERT: Only attention is needed for learning graph representations”, *arXiv preprint*, 2020.

Patent

- [38] **Haopeng Zhang**, Semih Yavuz, Wojciech Kryscinski, Kazuma Hashimoto, and Yingbo Zhou, *Systems and methods for abstractive document summarization with entity coverage control*, US Patent 11741142, Aug. 2023.

PROFESSIONAL SERVICE

• University Affiliation

- | | |
|---|----------------|
| – Cooperating Graduate Faculty, Department of Linguistics | 2025 - Present |
| – Affiliate Member, University of Hawaii Cancer Center | 2025 - Present |
| – Member, Hawaii Data Science Institute | 2024 - Present |
| – Member, NSF EPSCoR Change Hawaii (HI) Initiative | 2024 - Present |

• University Service

- | | |
|---|----------------|
| – Search Committee Member, Department of Information and Computer Sciences | 2025 - 2026 |
| – Graduate Studies Committee Member, ICS department | 2024 - Present |
| – Search Committee Member, Department of Oceanography | 2025 - 2026 |
| – Data Science Committee Member, Communication and Information Sciences (CIS) Program | 2024 - Present |

• Panel and Proposal Review Service

– Panelist, NSF CISE Program

2025, 2024

• **Conference and Workshop Service**

- Organizer, EMNLP 2025 Workshop on New Frontiers in Summarization (NewSumm) 2025
- Area Chair, ACL Rolling Review (ARR) 2026, 2025
- Area Chair, Conference on Neural Information Processing Systems (NeurIPS) 2026
- Area Chair, Conference on Empirical Methods in Natural Language Processing (EMNLP) 2025
- Area Chair, Annual Meeting of the Association for Computational Linguistics (ACL) 2025
- Minitrack Chair, Americas Conference on Information Systems (AMCIS) 2025

• **Journal Reviewer**

- IEEE/ACM Transactions on Audio, Speech, and Language Processing
- IEEE Transactions on Big Data
- IEEE Transactions on Computational Social Systems
- IEEE Transactions on Circuits and Systems for Video Technology
- IEEE Computer Magazine
- ACM Transactions on Embedded Computing Systems
- ACM Transactions on Social Computing
- Computer Speech and Language (Elsevier)
- Applied Soft Computing Journal
- Artificial Intelligence Review (Springer)
- Expert Systems with Applications (Elsevier)
- Engineering Applications of Artificial Intelligence (Elsevier)
- PeerJ Computer Science
- Frontiers in Artificial Intelligence

• **Conference Reviewer & Program Committee Member**

- Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL): 2025, 2024
- Annual Meeting of the Association for Computational Linguistics (ACL): 2024, 2023
- Conference on Empirical Methods in Natural Language Processing (EMNLP): 2024, 2023
- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD): 2024, 2023
- International Joint Conference on Neural Networks (IJCNN): 2024, 2023
- SIAM International Conference on Data Mining (SDM): 2024
- International Conference on Pattern Recognition (ICPR): 2024
- ACL Rolling Review (ARR): 2024, 2023
- Conference of the European Chapter of the Association for Computational Linguistics (EACL): 2023
- EMNLP 2023 Workshop on New Frontiers in Summarization (NewSumm): 2023
- ACL 2023 Workshop on Computational Approaches to Discourse (CODI): 2024
- AAAI Conference on Artificial Intelligence (AAAI): 2022, 2021

TEACHING EXPERIENCE

[†] denotes graduate-level courses.

- **Instructor, University of Hawaii at Manoa**

- ICS 691G[†] : Natural Language Processing and Large Language Models Fall 2025
- ICS 635[†] : Machine Learning Spring 2025
- ICS 435: Machine Learning Fundamentals Spring 2026, Spring 2025
- ICS 661[†] : Advanced Artificial Intelligence Fall 2024

- **Teaching Assistant, University of California, Davis**

- ECS 189G: Deep Learning Winter 2022

- **Teaching Assistant, Florida State University**

- COP 5725[†] : Advanced Database Systems Fall 2021, Spring 2021
- COP 4531: Complexity and Analysis of Data Structures and Algorithms Spring 2020
- COP 4710: Theory and Structure of Databases Fall 2020, Fall 2019

- **Teaching Assistant, Georgia Institute of Technology**

- ECE 3450: Semiconductor Devices Fall 2017, Spring 2017

MENTORSHIP

Underscore indicates my Ph.D. students.

Haohan Yuan

Fall 2024 - present

Ph.D. Student, University of Hawaii at Manoa
Research Area: Information Extraction, Summarization
Award: AI at Hawaii Data Science Institute Fellowship
Collaborated Publications: [6], [5], [2], [1]

Cade Kane

Fall 2025 - present

Ph.D. Student, University of Hawaii at Manoa
Research Area: Hawaiian language NLP

Shouju Wang

Fall 2025 - present

Ph.D. Student, University of Hawaii at Manoa
Research Area: Agentic AI privacy
Collaborated Publications: [34]

Mengqi Shi

Fall 2025 - present

Ph.D. Student, University of Hawaii at Manoa
Research Area: Video Summarization

Kaiying Lin

2024 - present

Assistant Research Fellow, Institute of Linguistics, Academia Sinica
Research Area: Low-resource Languages
Collaborated Publication: [8]

Zizhong Li

2023 - present

Ph.D. Student, University of California, Davis
Research Area: Retrieval-augmented Generation
Collaborated Publications: [3], [7], [11], [36]

Jolie Ching*Spring 2026 - present*

Master Student, University of Hawaii at Manoa
Master capstone project

Xiao Liu*2022 - 2024*

Ph.D. Candidate, University of California, Davis
Research Area: Text Summarization
Collaborated Publications: [14], [13], [15], [23], [16]

Samantha Mallari*2025 - 2026*

Master Student, University of Hawaii at Manoa
Master project: Benchmarking Vision-Language Models on Cross-View Camera Orientation Estimation